The Story So Far

Random phenomenon

$\downarrow$

Experiment

$\downarrow$

$\Omega,\ \omega,\ A$

$\downarrow$

$P(A)$

$\downarrow$

Conditional probability / independence / counting

$\downarrow$

$X : \Omega \to \mathbb{R}$

$\downarrow$

Distribution of X

$$\begin{cases} p_X(x), & \text{discrete} \\ f_X(x), & \text{continuous} \\ F_X(x), & \text{both} \end{cases}$$

$\downarrow$

Important numerical characteristics

$\downarrow$

$E[X], \quad \mathrm{Var}(X), \quad \sigma_X$

Properties of Variance

Suppose

$$Y = aX + b.$$

Then

$$E[Y] = aE[X] + b.$$

For the variance,

$$\operatorname{Var}[aX + b] = a^2 \operatorname{Var}[X].$$

Consequently,

$$\sigma_{aX+b} = |a|\sigma_X.$$

Moments of a Random Variable

The nth **moment** (or nth raw moment) of X is

$$E[X^n].$$

For example,

$$\begin{aligned} E[X] &: \text{ first moment,} \\ E[X^2] &: \text{ second moment.} \end{aligned}$$

The nth **central moment** is

$$E[(X - \mu_X)^n].$$

Therefore,

$$\operatorname{Var}[X] = E[(X - \mu_X)^2]$$

is the **second central moment** of X.

The terminology comes from mechanics, where moments describe how mass is distributed. Here, probability plays a role analogous to mass.

Where We Are Going

$$
\begin{array}{c}
X \\
\downarrow \\
\text{Distribution of } X \\
\downarrow \\
Y = g(X) \\
\downarrow \\
\text{Distribution of } Y \\
\downarrow \\
\text{Moments and other characterizations} \\
\downarrow \\
\text{Information carried by a random variable}
\end{array}
$$

Over the next two weeks, we're going to complete our study of a **single random variable** before moving on to multiple random variables.

Have a great Reading Week!

Enjoy the break, and I look forward to seeing you when we return.

